

Tetration

Anthony is just now learning basic math, how exciting! He first learns about addition

$$a + n = a + \underbrace{1 + 1 + \cdots + 1}_n,$$

then multiplication

$$a \times n = \underbrace{a + a + \cdots + a}_n,$$

exponentiation

$$a^n = \underbrace{a \times a \times \cdots \times a}_n.$$

and finally, *tetration*

$${}_n a = \underbrace{a^{a^{\cdots^a}}}_n.$$

Very quickly, Anthony becomes interested in infinite tetrations, namely

$${}^\infty a = a^{a^{\cdots}}.$$

Anthony wonders, given an arbitrary real number N , what is the solution to ${}^\infty a = N$? Unable to figure it out, Anthony has asked you to write a program to help him!

Here's a fun fact: A solution only exists for $\frac{1}{e} \leq N \leq e$.

Input

The first line of input contains one real number N , $0.36788 \leq N \leq 2.718281$.

Output

Output a single line containing a real number a , such that ${}^\infty a = N$. Your answer will be considered correct if its absolute or relative error doesn't exceed 10^{-5} .

Sample Input 1

Sample Output 1

2.000000



1.414214



Sample Input 2

Sample Output 2

1.000000



1.000000



Sample Input 3

Sample Output 3

1.500000



1.310371